

Nicolas Zilberstein

Houston, TX – nzilberstein@rice.edu

<https://sites.google.com/view/nzilberstein> – <https://github.com/nzilberstein>

EDUCATION

Rice University

Ph.D. in Electrical and Computer Engineering

Advisor: Prof. Santiago Segarra

Keywords: Inverse problems, Diffusion models, Sampling, Computational imaging

Houston, TX

2021 - 2026

Rice University

Master of Science in Electrical and Computer Engineering

Thesis: Annealed Langevin dynamics for massive MIMO communications

Houston, TX

2021 - 2023

University of Buenos Aires

B.S. in Electrical Engineering (with honors)

Thesis: Development of inference algorithms for UWB-GPR signals

Buenos Aires, Argentina

2013 - 2020

WORK EXPERIENCE

Google

Student Researcher at Computational Imaging Team (LUMA)

PIs: Kuldeep Purohit and Mojtaba Sahraee-Ardakan

Keywords: Diffusion models, Inference-time scaling, Inverse problems.

Mountain View, CA

May 2026 - August 2026

Center for Computational Neuroscience, Flatiron Institute

Summer research intern at the Computational Vision group

PIs: Eero Simoncelli and Florentin Guth

Keywords: Energy-based model, inverse problems, anisotropic diffusion models.

New York City, NY

May 2025 - August 2025

InterDigital

Research intern - AI for wireless

PIs: Akshay Malhotra and Shahab Hamidi-Rad

Keywords: Topological autoencoder, zero-shot stitching, latent space communication.

Los Altos, CA

May 2024 - August 2024

ExxonMobil

Data Engineer

Keywords: Front and Back-end, Cloud computing, Agile methodologies.

Buenos Aires, Argentina

2019 - 2021

School of Engineering, University of Buenos Aires

Research Assistant

PIs: Prof. Bruno Cernuschi-Frias, Prof. Juan Augusto Maya and Prof. Andres Altieri

Keywords: Particle filter, ground penetrating radar imaging, Green's function, sparse Bayesian learning.

Buenos Aires, Argentina

2018 - 2019

AWARDS

Grants

NVIDIA Academic Grant

2026

AI generated visuals: a solution for image demand in social science research

Center for Computational Insights on Inequality and Society (5000 USD)

Rice University

2025

Fellowships

Ken Kennedy – HPE Cray Graduate Fellowship

One-year graduate fellowships for outstanding graduate students at Rice University

Rice University

2024

UBACyT Scholarship (declined)

Scholarship for undergraduate students to develop a research project

University of Buenos Aires

2018

Honors

Top reviewer (Silver)

Selected as a top reviewer based on the area chairs rating

ICML
2026

Google PhD fellowship nominee

One of two nominees selected by the ECE department

Rice University
2025

Excellence in Research

Award from the ECE section of the Gulf Coast Undergraduate Research Symposium (GCURS)

Rice University
2020

Travel grants

ICML Travel Grant

2026

ICLR, AISTATS Travel Grant

2025

REA (Rice Engineering Alumni) Graduate Student Travel Grant

2025

PUBLICATIONS

Journals

[J1]

[J2]

[J3]

[J4]

Conference Papers

[C1]

[C2]

[C3]

[C4]

[C5]

[C6]

[C7]

[C8]

[C9]

[C10]

[C11]

[C12]

[C13]

Workshop Papers

[W1]

[W2]

[W3]

[W4]

Patents

1. “Systems and methods for statistics based and topology based interoperable AI/ML model monitoring”
Akshay Malhotra, Yugeswar Deenoo, Nicolas Zilberstein, Shahab Hamidi-Rad, Mohamed Salah Ibrahim
InterDigital, US Patent Ref. 18889096

2024

Preprints

[P1]

*Equal contribution

TEACHING EXPERIENCE

Department of Electrical and Computer Engineering, Rice University Houston, TX
Teaching Assistant

- ELEC 573 Network Science and Analytics, *Graduate course* Fall 2022, 2024
- ELEC 301 Signals and Systems, *Undergraduate course* Spring 2022, 2023, 2024, 2025

Graduate school of Computer Engineering, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- Deep Learning Fall 2021

Department of Physics, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- 82.02: Physics II (Electromagnetism) Spring 2020, Fall 2020

Department of Electronic Engineering, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- 86.55: Detection and Estimation theory Spring 2018, 2019

PROFESSIONAL ACTIVITIES

Invited talks

From Guided Sampling to Learned Solvers: Diffusion Models for Inverse Problems Dec 2025
Universidad de la Republica Uruguay

From Guided Sampling to Learned Solvers: Diffusion Models for Inverse Problems Nov 2025
Yusu Wang's group at *UCSD*

Towards fast samplers for solving inverse problems using diffusion models June 2024
Pendulum Systems

Organization

Special Session: Generative AI for Graphs Asilomar 2026
Organizer and Technical Chair

Workshops

Advances of Approximate Bayesian Inference (AABI) 2025 Singapore
Repulsive Latent Score Distillation for Solving Inverse Problems [?] presented in the fast track

Khipu AI 2025 Santiago, Chile
The largest meet-up gathering students, researchers, and innovators in the field of Artificial Intelligence and Machine Learning across Latin America

LOGML Summer School 2024 London, UK
Workshop that aims to bring together mathematicians and computer scientists to work together on a variety of problems at the intersection of geometry and machine learning

Graph Signal Processing Workshop 2023 Oxford, UK
Workshop on graph signal processing. Poster presentation.

Reviewer

Journals: IEEE Comm. Letter; IEEE Trans. Cogn. Commun.; IEEE Trans. Signal Process.; IEEE Trans. Commun.
Conferences: ICCASP, EUSIPCO, ICC, SAM, CAMSAP, NeurIPS (2024, 2025), ICLR (2026), ICML (2026)

Memberships

IEEE Member 2021 - Present
IEEE Signal Processing Society Member 2021 - Present

STUDENT ADVISEMENT

Graduate

Ali Azizpour

Undergraduate

Haoming Chen

Summer 2025

Uros Males, Mila Bokan

Summer 2023

OTHER SKILLS

Languages: Spanish (native), English (fluent)

COMMUNITY SERVICES

Evaluator at Gulf Coast Undergraduate Research Symposium (GCURS)

2024, 2025

Mentor of incoming ECE PhD students - Data science track

2024

Organizer, LatAm RICE Networks, Robots, and ML Minisymposium, Rice Univ.

2021

Member of the Council of the Faculty of Engineering, Univ. of Buenos Aires

2019 - 2021

Member, Advisory Board, Dept. of Electronics, Univ. of Buenos Aires

2018 - 2021

Member, Curricular Committee, Electronic Engineering Degree, Univ. of Buenos Aires

2017 - 2021